

Sounbum Song

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WORK & LEADERSHIP EXPERIENCE

HeadSpin

Riverside, CA (Remote)

Senior Software Engineer

June 2022 - Present

- Act as a Tech Lead in the Platform team, collaborating with the Product team to propose, validate, and determine backend architectures that align technical strategy with product roadmaps.
- Improve and modernize the internal messaging system originally based on a relational DB by implementing Kafka. Grafana Alloy reads messages from Kafka and stores in Grafana Loki, and then messages are visualized on Grafana dashboards.
 - Improve scalability and availability by reducing size of data to 1/7 after discarding indexes used in the relational DB.
 - Enhance visibility by increasing the length of retention period from days to years and implementing graphs on Grafana dashboards.
 - Optimize storage costs by automating historical data migration to S3 Glacier.
- Optimize internal data structure by implementing cache based on expiring HashMap in Redis, which eliminates bottlenecks in the main DB.
 - This speeds up function calls by 32% on average.
- Design a light-weight admin API service leveraging API gateway and Lambda.

HeadSpin

Palo Alto, CA

Software engineer

February 2020 - May 2022

- Designed a reverse proxy to support customers with limited network access to use HeadSpin services with a single domain and single port.
 - A light-weight service written in Golang that relays ~1,000 HTTP and WS requests per second.
- Architected an internal message and monitoring system to handle errors proactively and to audit usage by leveraging Sqlite as cache on edge machines and Mysql as the main DB.
 - This service handles ~50 million events per day.
 - Internal facing UI is built using React and Material UI.

HeadSpin

Palo Alto, CA

Intern

June 2018 - August 2018

- Developed APIs that allow customers to integrate HeadSpin services into Slack to monitor and get alerts about test status.
- Developed a Jenkins Plugin to run HeadSpin automations in customers own test pipelines.

SAIC

Seoul, Korea (Remote)

Lead Developer

June 2018 - February 2020

- Developed websites and Windows executables leveraging HTML, CSS, JavaScript, Python, Django and MySQL
- Cryptocurrency Auto Arbitrage Trader
 - Designed a profit-making algorithm by using price differences between cryptocurrency exchanges.
 - Leveraged multi-threading and Asyncio to improve performance and accuracy and to reduce time (~40%) waiting for response of data from RestAPIs and Websockets provided by multiple cryptocurrency exchanges.

Freelancer

Seoul, Korea

Full-stack Developer

August 2017 – June 2018

- Maintained and developed full-stack applications leveraging Python and API server with Flask and SQLite
- CRM
 - Designed and developed client, server and database for organizing data from users and accessing the data from multiple locations (~100) with administrative mode to control clients
 - Leveraged RestAPI to communicate between clients and server to create, update and delete data
- Auto Product Management System

- Designed and developed client, server and database for registering, updating and deleting products to multiple online shopping malls.
- Leveraged Selenium to automate tasks such as updating products based on database to online shopping malls that do not provide APIs
- Reduced redundant tasks such as processing same product among clients by implementing a server which distributes tasks and data to multiple clients (~50)

Republic of Korea Cyber Command

Seoul, Korea

Information Security Application Developer

May 2015 – August 2017

- Maintained and developed malware threat monitoring system implementing database replication and procedures to handle big data (100 million~)
- Distributed system using XenServer collected malware information and analyzed severity and threat level on private networks in the military and public networks of Korea
- Analyzed malware using reversing and debugging techniques with IDA Pro and Immunity Debugger
- Developed websites that provide a view of monitoring result and saving and retrieving malware analyzation data

PERSONAL PROJECTS

Turtle Trading Agent

- System trading service leveraging “turtle” algorithm
- Leverages open APIs from brokerage to read real time ohlvc and orderbook data.
- Splits systems into microservices to improve catching signals and submitting orders for multiple stocks.
 - leverages Redis queue for quick communications between services.
- Training Informer, a transformer model for long time series data, with real time data.

Auto Stock Trader

- Implemented an artificial intelligence algorithm - SVM, to predict price and trend of a stock using technical indicators calculated from public data using Python and OpenCV
- Improved performance of calculating technical indicators by using NumPy instead of simple loops
- Learned the basics of selecting and testing data to train an AI

EDUCATION

University of Arizona

Tucson, Arizona

Bachelor of Science in Computer Science

August 2017 - December 2019

Bachelor of Science in Mathematics

- **GPA:** 3.841 / 4.0;

SKILLS, ACTIVITIES & INTERESTS

Technical Skills:

- **Language:** Python, Go, Java
- **Infrastructure/Data:** Kafka, Redis, MySQL, SQLite, Docker, Grafana(Alloy, Loki, Dashboards)

Languages: English, Korean

Activities: SecurityPlus Union Academy, WhiteHat Contest